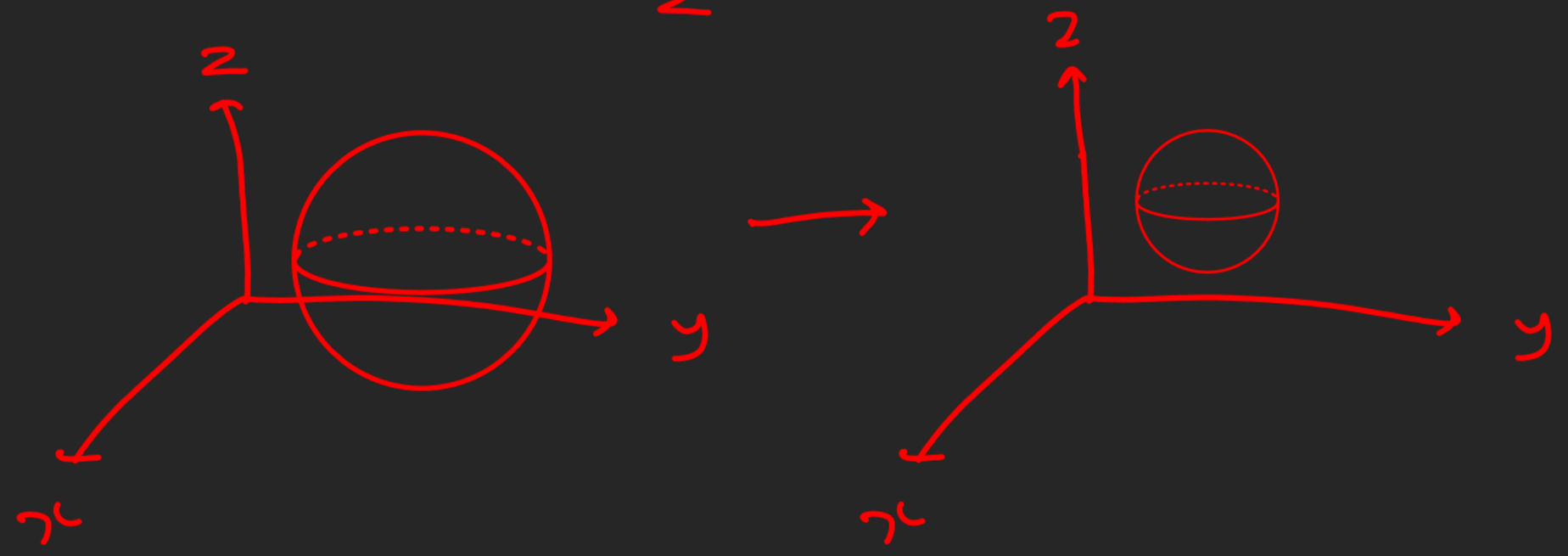
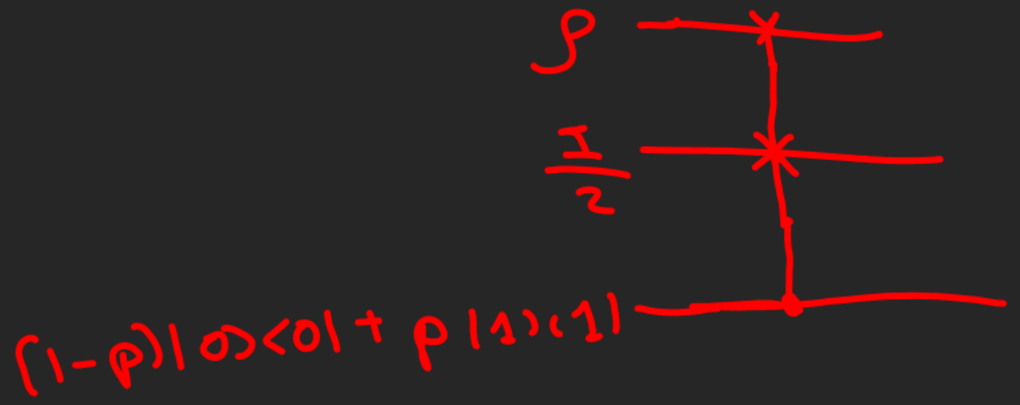
Depolarising Channel

$$\Omega_p(\rho) = p \cdot \frac{I}{2} + (1-p) \cdot \rho$$





$$\frac{I}{2} = \frac{\rho + X\rho X + Y\rho Y + Z\rho Z}{4}$$

$$\Rightarrow p(\rho) = (1-p')\rho + \frac{p'}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

$$\hookrightarrow p' = \frac{3p}{4}$$

$$F_e = \langle\phi^+|\rho|\phi^+\rangle = 1-p$$ $\rightarrow$ we can interpret it as probability of no error measured

$$\rho(\vec{P}) = \frac{I + \vec{P}\cdot\vec{\sigma}}{2}$$

$\vec{P} \rightarrow$ spin Polarisation

$$\rho' = \left(1 - \frac{4p}{3}\right)\rho + \frac{4p}{3} I = \rho(\vec{P}')$$

$$\vec{P}' = \left(1 - \frac{4p}{3}\right)\vec{P}$$

$\rightarrow$ For $p \leq \frac{3}{4}$ $\rightarrow$ Spin polarisation shrinks by a factor of $\left(1 - \frac{4p}{3}\right)$

In general, for d-dimensional quantum system

$$\Omega_p(\rho) = p \cdot \frac{I}{d} + (1-p)\rho$$